Lesson 2: Variables in Loops: Spirals and Growing Patterns

Module 1 | 45 minutes | Variables inside loops | Grades 3–5 · ages 8–10

Name: _____ Class: _____ Date: _____

Practise what you learned today. Read each task carefully.

1. Key words

Match each word to what it means. Write the letter.

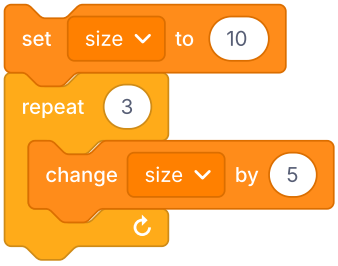
a. variable ___ A. A line that winds around, getting further from the middle.
b. reporter ___ B. Set a variable's starting value.
c. initialise ___ C. A round block that gives a value, like a variable.
d. spiral ___ D. A named box that stores a value.

2. Plan your code

Before you build it: Draw a square spiral: set length to 10, then repeat 20 times: move length steps, turn 90, and change length by 10. Write or draw the blocks you will use.

3. Trace the variable

What is the value of size at the end? Circle one.



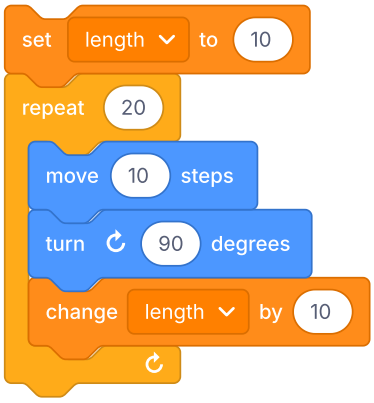
a. A. 25
b. B. 15
c. C. 30

4. Plan your code

Before you build it: Make Pip count from 1 to 10 out loud: say each number for half a second. Write or draw the blocks you will use.

5. Debug it

Mia's spiral only draws the same short line again and again. What's wrong? Circle one.



a. A. The move block uses the number 10 instead of the length variable
b. B. The repeat should be 10
c. C. The turn should be 45

How did it go? Colour a face: 😀 I can do it · 🙂 Nearly there · 🤔 I need help

☐ I can set a variable before a loop and change it inside.
☐ I can use a variable reporter inside a block.
☐ I can draw a growing spiral.